$$F(R,P) = \frac{(1+\beta^2)PR}{\beta^2 P + R}$$

1. $\beta = 1$, i.e., P and R equally important, we obtain

$F(R,P) = \frac{2PR}{P+R} = \frac{2}{\left\{\frac{1}{R}+\frac{1}{P}\right\}}$ = the harmonic mean <= geometric mean <= arithmetic mean

2. $\beta^2 = 0.5$

$$F(R,P) = \frac{3/2\,PR}{1/2\,P+R} = \frac{3PR}{P+2R}$$

3. $\beta = 0.5$

$$F(R,P) = \frac{(1+0.25)PR}{0.25\ \ P+R} = \frac{1.25\,P\,R}{0.25\,P+R} = \frac{1.25\,PR}{0.25(P+4R)} = 5\ \frac{PR}{P+4R}$$

4. $\beta = 0$ $\qquad F(R,P) = \frac{PR}{R} = R$

5. $\lim_{\beta \to \infty} F(R,P) = \frac{PR}{R} = P$